

What is optimization?

Optimization is a subject that is widely and increasingly used in science, engineering, economics, management, industry and on other areas. It deals with selecting the best of many possible decisions, of real life environment, constructing computational methods for finding optimal solutions, exploring theoretical properties and studying computational performance of numerical computational algorithms implemented based on the optimization problems.

Constrained Optimization Problem

Consider the optimization problem

minimize $f(x)$

subject to $x \in \Omega$

The function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ that we have to minimize is a real-valued function and is called the objective function or cost function. The vector x is an n -vector of independent variables. The set Ω is a subset of $\mathbb{R}^n$, called the ~~constrained~~ constraint or feasible set. The above optimization problem is called constrained optimization problem.

Unconstrained Optimization Problem

If $\mathcal{R} = \mathbb{R}^n$, then the optimization problem is called unconstrained optimization problem.

Convex Set

The line segment between two point $u, v \in \mathbb{R}^n$ is the set $\{w \in \mathbb{R}^n : w = \alpha u + (1-\alpha)v, \alpha \in [0,1]\}$. A set $\theta \subset \mathbb{R}^n$ is called convex if for all $u, v \in \theta$, the line segment between u and v is in θ .

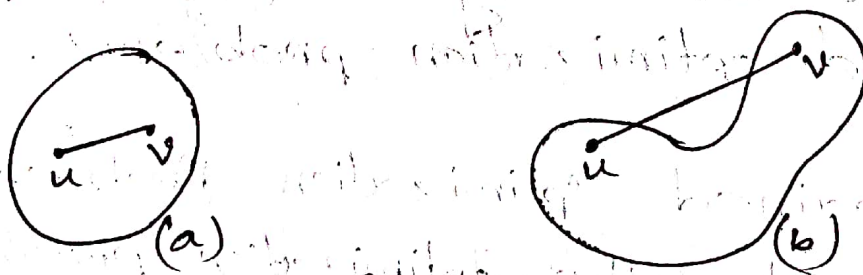


Figure: 1

Figure 1(a) gives example of convex set whereas Figure 1(b) gives example of a set that is not convex.

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Jacobian Matrix / Derivative Matrix

Consider a differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. Let

$$x = (x_1, x_2, \dots, x_n)^T$$

If ~~and~~ $f(x) = [f_1(x), f_2(x), \dots, f_m(x)]^T$

then

$$\frac{\partial f}{\partial x_j}(x_0) = \left[\frac{\partial f_1}{\partial x_j}(x_0), \frac{\partial f_2}{\partial x_j}(x_0), \dots, \frac{\partial f_m}{\partial x_j}(x_0) \right]^T$$

Now

$$\left[\frac{\partial f}{\partial x_1}(x_0), \frac{\partial f}{\partial x_2}(x_0), \dots, \frac{\partial f}{\partial x_n}(x_0) \right]$$

$$= \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \frac{\partial f_1}{\partial x_2}(x_0) & \dots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \frac{\partial f_2}{\partial x_2}(x_0) & \dots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \frac{\partial f_m}{\partial x_2}(x_0) & \dots & \frac{\partial f_m}{\partial x_n}(x_0) \end{bmatrix}$$

$$= L$$

The matrix L is called the Jacobian matrix or derivative matrix of f at x_0 and is denoted by $Df(x_0)$.

Gradient of f

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, then the function ∇f defined by

$$\nabla f(x) = \left[\frac{\partial f}{\partial x_1}(x) \quad \frac{\partial f}{\partial x_2}(x) \quad \dots \quad \frac{\partial f}{\partial x_n}(x) \right]^T$$

$$= Df(x)^T$$

is called the gradient of f .

Hessian Matrix

Given $f: \mathbb{R}^n \rightarrow \mathbb{R}$. If ∇f is differentiable and f is twice differentiable, we write the derivative of ∇f

$$D^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \dots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_n \partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

The matrix $D^2f(x)$ is called the Hessian matrix of f at x and is also denoted by $F(x)$.

Example 6.1 ; Page 94

Let $f(x_1, x_2) = 5x_1 + 8x_2 + x_1x_2 - x_1^2 - 2x_2^2$.

Find ∇f , Df , $F(x)$.

Note:-

∇f = column vector
 $Df, F(x)$ = Matrix

Solution:- Given that

$$f(x_1, x_2) = 5x_1 + 8x_2 + x_1x_2 - x_1^2 - 2x_2^2$$

Now $\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x) \\ \frac{\partial f}{\partial x_2}(x) \end{bmatrix}$

$$= \begin{bmatrix} 5 + x_2 - 2x_1 \\ 8 + x_1 - 4x_2 \end{bmatrix}$$

$$Df(x) = (\nabla f(x))^T = [5 + x_2 - 2x_1, 8 + x_1 - 4x_2]$$

$$F(x) = D^2 f(x) = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) \\ \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \\ 1 & -4 \end{bmatrix}$$

Feasible Direction

A vector $d \in \mathbb{R}^n$, $d \neq 0$, is called a feasible direction at $x \in \Omega$ if there exists $\alpha_0 > 0$ such that $x + \alpha d \in \Omega$ for all $\alpha \in [0, \alpha_0]$

Local minimizer

Suppose that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a real valued function, defined on some set $\Omega \in \mathbb{R}^n$. A point $x^* \in \Omega$ is a local minimizer of f over Ω if there exists $\epsilon > 0$ such that $f(x) \geq f(x^*)$ for all $x \in \Omega \setminus \{x^*\}$ and $\|x - x^*\| < \epsilon$.

Note:- $\|x - x^*\| < \epsilon$ i.e. x & x^* points are too close.
 → Local minimizer - a point x^* $\nabla f(x^*) = 0$
 → Global " " " " " " " " " " " "

Global minimizer

Suppose that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a real valued function defined on some set $\Omega \in \mathbb{R}^n$.
 A point $x^* \in \Omega$ is a global minimizer of f over Ω if $f(x) \geq f(x^*)$ for all $x \in \Omega \setminus \{x^*\}$.

Directional derivative (अवकल)

$$\frac{\partial f(x)}{\partial d} = \nabla f(x)^T d = d^T \nabla f(x)$$

Problem:- Define $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ by $f(x) = x_1 x_2 x_3$ and let $d = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right]^T$. Find $\frac{\partial f}{\partial d}(x)$.

Solution:- Given that $f(x) = x_1 x_2 x_3$ and $d = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right]^T$.

$$\text{Now } \nabla f(x) = \begin{bmatrix} x_2 x_3 \\ x_1 x_3 \\ x_1 x_2 \end{bmatrix}$$

The directional derivative of f in the direction d is

$$\begin{aligned} \frac{\partial f}{\partial d}(x) &= \nabla f(x)^T d = [x_2 x_3, x_1 x_3, x_1 x_2] \begin{bmatrix} 1/2 \\ 1/2 \\ 1/\sqrt{2} \end{bmatrix} \\ &= \frac{x_2 x_3 + x_1 x_3 + \sqrt{2} x_1 x_2}{2} \end{aligned}$$

Theorem 6.1: FONC

Statement:- Let Ω be a subset of $\mathbb{R}^n$ and $f \in C^1$ a real valued function on Ω . If x^* is a local minimizer of f over Ω , then for any feasible direction d at x^* , we have

$$d^T \nabla f(x^*) \geq 0.$$

Proof:- We define

$$x(\alpha) = x^* + \alpha d \in \Omega$$

then $x(0) = x^*$. Now define the composite function $\phi(\alpha) = f(x(\alpha))$. Then by Taylor's theorem

$$\begin{aligned} f(x^* + \alpha d) - f(x^*) &= \phi(\alpha) - \phi(0) \\ &= \phi'(\alpha) \alpha + o(\alpha) \end{aligned}$$

$$= \alpha d^T \nabla f(x(0)) + o(\alpha)$$

$$\text{or, } f(x^* + \alpha d) = f(x^*) + \alpha d^T \nabla f(x(0)) + o(\alpha)$$

where $\alpha \geq 0$. Thus if $\phi(\alpha) \geq \phi(0)$ that is

$f(x^* + \alpha d) \geq f(x^*)$ for small values of $\alpha > 0$, then

$$d^T \nabla f(x^*) \geq 0.$$

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Problem 6.3:- Consider the problem

$$\text{minimize } x_1^2 + 0.5 x_2^2 + 3x_2 + 4.5$$

subject to $x_1, x_2 \geq 0$

Is the first order necessary condition (FONC) for a local minimizer satisfied at
 i) $x = [1, 3]^T$ ii) $[0, 3]^T$ iii) $[1, 0]^T$ iv) $[0, 0]^T$.

→ FONC $d^T \nabla f(x^*) \geq 0$. For interior case $\nabla f(x^*) = 0$

Solution:-

i) At $x = [1, 3]^T$, we have

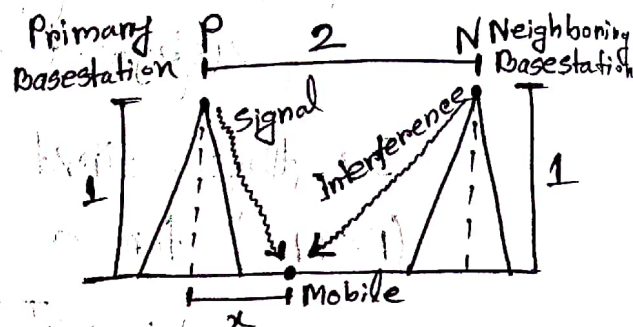
$$\begin{aligned} \nabla f(x) &= [2x_1, x_2 + 3]^T \\ &= [2, 6]^T \end{aligned}$$

The point $x = [1, 3]^T$ is an interior point of $\Omega = \{x : x_1 \geq 0, x_2 \geq 0\}$. Hence the FONC requires $\nabla f(x) = 0$. The point $x = [1, 3]^T$ does not satisfy the FONC for a local minimizer.

ii) iii) iv) From book page - 79:

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Problem:- There are two base station antennas P and N. Both antennas are transferring signals to the mobile user at equal power.



The power of the reduced signal as measured by the mobile to the reciprocal of the squared distance from the associated antenna P. Find the position of the mobile that minimizes the signal to interference ratio, which is the ratio of the reduced signal power from P to N.

Solution:-

From the given figure

Power of signal $1+x^2$

Power of interference $1+(2-x)^2$

Figure

Therefore, the signal-to-interference ratio is,

$$f(x) = \frac{1+x^{\sqrt{2}}}{1+(2-x)^{\sqrt{2}}}$$

We have

$$f'(x) = \frac{\{1+(2-x)^{\sqrt{2}}\} \cdot 2x - (1+x^{\sqrt{2}})(2x-4)}{\{1+(2-x)^{\sqrt{2}}\}^{\sqrt{2}}}$$

$$= \frac{1-4(x^{\sqrt{2}}-2x-1)}{\{1+(2-x)^{\sqrt{2}}\}^{\sqrt{2}}}$$

By the FONC, at the optimal position

$$f'(x) = 0$$

$$\Rightarrow x^{\sqrt{2}} - 2x - 1 = 0$$

$$\Rightarrow x = 1 \pm \sqrt{2}$$

Substitute these values in the function we have to find the minimum value.

We see that $x = 1 - \sqrt{2}$, the value of the function is minimum.

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State and prove SOSC

Statement:- Let $\Omega \in \mathbb{R}^n$, $f \in C^2$ a function on Ω , x^* a local minimizer of f over Ω and d a feasible direction at x^* . If $d^T \nabla f(x^*) = 0$, then

$$d^T F(x^*) d \geq 0$$

where F is the Hessian of f .

Proof:- We prove the result by contradiction. Suppose that there is a feasible direction d at x^* such that $d^T \nabla f(x^*) = 0$ and $d^T F(x^*) d < 0$. Let $x(\alpha) = x^* + \alpha d$ and define the composite function $\phi(\alpha) = f(x^* + \alpha d) = f(x(\alpha))$. Then by Taylor's theorem

$$\phi(\alpha) = \phi(0) + \phi'(0) \frac{\alpha^r}{2} + o(\alpha^r)$$

where by assumption $\phi'(0) = d^T \nabla f(x^*) = 0$ and $\phi''(0) = d^T F(x^*) d < 0$. For small α

$$\phi(\alpha) - \phi(0) = \phi''(0) \frac{\alpha^r}{2} + o(\alpha^r) < 0$$

$$\Rightarrow f(x^* + \alpha d) - f(x^*) < 0$$

$$\Rightarrow f(x^* + \alpha d) < f(x^*)$$

which contradicts the assumption that x^* is a local minimizer. Thus

$$\phi''(0) = d^T F(x^*) d \geq 0.$$

hence x^* is a local minimizer

Use the Golden Section Search Method to find the value of x that minimize

$$f(x) = x^4 - 14x^3 + 60x^2 - 70x$$

in the range $[0, 2]$. Locate the value of x to within a range of (0.3) .

Solution: Given that

$$f(x) = x^4 - 14x^3 + 60x^2 - 70x$$

Here four stages of reduction will do i.e. $N=4$ and $p=0.382$. $[a_0, b_0] = [0, 2]$.

We evaluate f at two intermediate points a_1 and b_1 . We have

$$a_1 = a_0 + p(b_0 - a_0) = 0.7639$$

$$b_1 = a_0 + (1-p)(b_0 - a_0) = 1.236$$

Here

$$f(a_1) = -24.36$$

$$f(b_1) = -18.96$$

Thus $f(a_1) < f(b_1)$ and so the interval is reduced to

$$[a_0, b_1] = [0, 1.236]$$

Now we choose $b_2 = a_1$ and

$$a_2 = a_0 + p(b_1 - a_0) = 0.4721$$

We have

$$f(a_2) = -21.10$$

$$f(b_2) = -24.36$$

Since $f(b_2) < f(a_2)$, the interval is reduced to

$$[a_2, b_1] = [0.4721, 1.236]$$

We choose $a_3 = b_2$ and

$$b_3 = a_2 + (1-p)(b_1 - a_2) = 0.9443$$

We have

$$f(a_3) = -24.36$$

$$f(b_3) = -23.59$$

Since $f(b_3) > f(a_3)$, the interval is reduced to

$$[a_2, b_3] = [0.4721, 0.9443]$$

We choose $b_4 = a_3$ and

$$a_4 = a_2 + p(b_3 - a_2) = 0.6525$$

Thus

$$f(a_4) = -23.84$$

$$f(b_4) = -24.36$$

Since $f(a_4) > f(b_4)$, thus the value of x that minimizes f is located in the interval

$$[a_4, b_3] = [0.6525, 0.9443]$$

and the value is

$$b_3 - a_4 = 0.292 < 0.3.$$

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